ROADMAP: From Zero to Correctly Understanding Chatbots/AI (based on papers & evidence)

The goal of this roadmap is to separate three things that people often mix up:

- "Random guessing"
- "Estimating a probability distribution" (estimate a distribution using data + a model)
- "Text generation" (generation = the model computes a distribution + a decoding algorithm chooses tokens)

The through-line of the roadmap: a chatbot/LLM doesn't "have intuition" or "guess randomly." It runs a computation graph to produce a probability distribution over the next token; then decoding decides which token gets emitted. The strongest evidence sits in papers on decoding and on analyzing attention/forward-pass mechanisms. (arXiv)

PHASE 0: Math foundations — Probability — Information

Purpose: understand "probability as extended logic," and understand entropy/cross-entropy as measurement quantities, not "chaos."

P0.1 Basic probability

- Edwin T. Jaynes, *Probability Theory: The Logic of Science* (book; widely circulated edition 2003; drafts circulated earlier). Focus on early chapters (plausible reasoning, rules of probability). (Bayes Institute)
- Key insight: probability is a way to encode "degree of plausibility" under uncertainty, not a game of luck.

P0.2 Information Theory

- Claude E. Shannon, "A Mathematical Theory of Communication" (Bell System Technical Journal, 1948). (Harvard Math People)
- Cover & Thomas, *Elements of Information Theory* (2nd ed., widely cited; Wiley). (USTC)
- Key insight: entropy/cross-entropy measure "surprise" and model quality in the statistical sense.

Checkpoint (Phase 0)

- You can explain: how “having a probability distribution” differs from “guessing randomly” (because the distribution is constrained by data and by optimization rules / information theory).

PHASE 1: Classical language modeling (pre–deep learning)

Purpose: see that “predict next token” is an old statistical problem, not a new invention.

P1.1 N-gram models

- Chen & Goodman, “An Empirical Study of Smoothing Techniques for Language Modeling” (appears as paper/tech report; arXiv version from 1996; widely treated as a classic on smoothing). (arXiv)
- Key insight: “next-token prediction” = counting + Markov assumption + smoothing to avoid zero probability.

P1.2 Statistical thinking in language models

- Frederick Jelinek, *Statistical Methods for Speech Recognition* (1997), a classic source for the statistical tradition (LM tightly linked with speech). (Semantic Scholar)
- Key insight: classical LM is already probability optimization + data, with no “intuition.”

Checkpoint (Phase 1)

- You can explain why “predict next word” is a probabilistic task, and why smoothing changes LM quality.

PHASE 2: Neural language models — from counting to representations

Purpose: understand “representation learning” instead of pure counting.

P2.1 Embeddings and neural probabilistic LM

- Bengio et al., “A Neural Probabilistic Language Model” (JMLR, 2003). (JMLR)
- Key insight: words become points in a continuous space; optimizing prediction makes semantic structure “emerge.”

P2.2 Word2Vec

- Mikolov et al., “Efficient Estimation of Word Representations in Vector Space” (2013). (Read as a minimal paper about objective and geometric consequences.) (arXiv)
- Key insight: a simple objective can yield geometry that encodes relations.

Checkpoint (Phase 2)

- You can explain: “learning to predict” is not “memorizing verbatim,” but compressing the data distribution into parameters.

PHASE 3: Attention & Transformer — the backbone of modern chatbots

Purpose: see clearly that attention is computation (weighted lookup), and Transformer is a chain of matrix operations.

P3.1 Attention mechanism

- Bahdanau, Cho, Bengio, “Neural Machine Translation by Jointly Learning to Align and Translate” (2014; often associated with ICLR 2015). (arXiv)
- Key insight: attention scores come from computations on query/key, not “mental focus.”

P3.2 Transformer

- Vaswani et al., “Attention Is All You Need” (2017). (arXiv)
- Focus: scaled dot-product attention, multi-head attention, the architecture diagram.

Checkpoint (Phase 3)

- You can describe the forward pass at the level of “blocks of operations” (projections, attention, FFN, residual, layer norm).

PHASE 4: Pre-training — what “learning” actually is

Purpose: clearly separate “training has feedback” vs “inference just computes with fixed weights.”

P4.1 GPT-style pretraining

- Radford et al., “Improving Language Understanding by Generative Pre-Training” (GPT-1, 2018). (OpenAI CDN)
- Radford et al., “Language Models are Unsupervised Multitask Learners” (GPT-2, 2019). (OpenAI CDN)

- Key insight: a next-token objective can transfer to many tasks.

P4.2 Scaling laws

- Kaplan et al., “Scaling Laws for Neural Language Models” (2020). (arXiv)
- Key insight: loss/performance follows predictable laws with scale (data/compute/params).

Checkpoint (Phase 4)

- You can explain: training optimizes cross-entropy via gradient methods; inference does not update weights (but still “estimates” next-token distributions in the statistical sense).

PHASE 5: Emergence & in-context learning

Purpose: explain why it “seems like understanding” without assuming “consciousness.”

P5.1 Few-shot / in-context learning

- Brown et al., “Language Models are Few-Shot Learners” (GPT-3, 2020). (arXiv)
- Garg et al., “What Can Transformers Learn In-Context? A Case Study of Simple Function Classes” (ICLR 2022). (arXiv)
- Key insight: ICL is “computation in the forward pass” (no backprop during chat).

P5.2 Emergent abilities

- Wei et al., “Emergent Abilities of Large Language Models” (TMLR, 2022). (Proceedings NeurIPS)
- Key insight: “sudden emergence” can be threshold / phase-transition effects with scale, not magic.

Checkpoint (Phase 5)

- You can articulate: “capability” emerges from optimization + architecture + scale.

PHASE 6: Generation & decoding — the key to killing the “AI guesses randomly” myth

Purpose: prove that “the model computes a distribution” and “decoding chooses tokens” are two different layers; changing decoding can change output quality/style drastically even if the model is unchanged.

P6.1 Nucleus sampling and text degeneration

- Holtzman et al., “The Curious Case of Neural Text Degeneration” (ICLR 2020; arXiv 2019/2020). (arXiv)
- Evidence to retain: same model, different decoding → strongly different outputs.

P6.2 Decoding taxonomy overview

- Wiher, Meister, Cotterell, “On Decoding Strategies for Neural Text Generators” (TACL, 2022). (ACL Anthology)

Checkpoint (Phase 6)

- You can do a demo:
  - temperature = 0 / greedy → almost reproducible
  - different top-p/top-k/temperature → “controlled stochasticity,” not “random guessing”

PHASE 7: Inference mechanics — opening the “black box” of the forward pass

Purpose: see “circuits” and attention patterns as traceable computational mechanisms.

P7.1 Transformer circuits (mechanistic interpretability)

- Elhage et al., “A Mathematical Framework for Transformer Circuits” (Transformer Circuits / Anthropic). (OpenReview)
- Olsson et al., “In-context Learning and Induction Heads” (2022). (arXiv)
- Key insight: some heads play specific roles; they form through training dynamics.

P7.2 Long context and attention sinks

- Xiao et al., “Efficient Streaming Language Models with Attention Sinks” (appears 2023; associated with ICLR 2024). (arXiv)
- Key insight: early tokens can attract attention to stabilize computation; behavior can be “computational structure,” not pure semantics.

Checkpoint (Phase 7)

- You can explain: many LLM behaviors are consequences of attention/KV-cache/numerical stability, without attributing “intent.”

PHASE 8: Training dynamics — learning vs inference as a feedback loop

Purpose: training is an optimization loop (objective + gradient), inference is just forward computation with fixed weights.

P8.1 Optimization

- Ruder, “An Overview of Gradient Descent Optimization Algorithms” (2016). (arXiv)

P8.2 Loss landscape

- Li et al., “Visualizing the Loss Landscape of Neural Nets” (NeurIPS 2018). (NIPS Papers)

P8.3 Generalization & memorization

- Zhang et al., “Understanding Deep Learning Requires Rethinking Generalization” (ICLR 2017). (arXiv)

Checkpoint (Phase 8)

- You can describe: “learning” is moving in a loss landscape; “text generation” is reading a distribution and applying decoding.

PHASE 9: Limitations & failures — hallucination and reasoning limits

Purpose: instead of saying “AI makes things up,” treat it as structured consequences of data/architecture/decoding.

P9.1 Hallucination taxonomy

- Ji et al., “A Survey of Hallucination in Natural Language Generation” (ACM Computing Surveys, 2023). (arXiv)
- McKenna et al., “Sources of Hallucination in Large Language Models” (2023). (arXiv)

P9.2 Compositional reasoning limits

- Dziri et al., “Faith and Fate: Limits of Transformers on Compositionality” (NeurIPS 2023). (NIPS Papers)

Checkpoint (Phase 9)

- You can run a test: even with temp=0 the model can hallucinate in a repeatable way → that's a reproducible artifact, not meaningless randomness.

PHASE 10: Theoretical foundations — why next-token prediction is not "trivial"

Purpose: see the theoretical power of autoregressive learning.

P10.1 Universal learners

- Malach, "Auto-Regressive Next-Token Predictors are Universal Learners" (2023). (arXiv)

P10.2 (Suggested replacement for "Sanh et al.")

- If you want the "universality / computational simulation" branch for transformers/AR models, pick papers with clean citations about expressivity/computation (depending on whether you want theory-only, transformer-specific, or AR-specific), instead of keeping an entry whose title/venue can't be verified.

Checkpoint (Phase 10)

- You can say: a "simple objective" does not imply "simple capability."

PHASE 11: Alignment & RLHF — from base model to chatbot

Purpose: a chatbot differs from a "base LM" via the alignment pipeline, not because it "has a personality."

P11.1 Instruction tuning

- Wei et al., "Finetuned Language Models are Zero-Shot Learners" (FLAN, 2022). (arXiv)

P11.2 RLHF / preference optimization

- Ouyang et al., "Training language models to follow instructions with human feedback" (InstructGPT, 2022). (arXiv)
- Bai et al., "Constitutional AI: Harmlessness from AI Feedback" (2022). (arXiv)

Checkpoint (Phase 11)

- You can describe the pipeline: SFT → reward model → policy optimization (and understand it's still optimization).

PHASE 12: Production systems — real chatbots run fast via compute optimization

Purpose: deployment doesn't add "magic," it mostly optimizes compute/memory/throughput.

P12.1 Inference optimization

- Shazeer, "Fast Transformer Decoding: One Write-Head is All You Need" (2019). (arXiv)
- Dao et al., "FlashAttention: Fast and Memory-Efficient Exact Attention" (NeurIPS 2022). (ACM Digital Library)

P12.2 Serving & scaling inference

- Pope et al., "Efficiently Scaling Transformer Inference" (2022). (arXiv)

Checkpoint (Phase 12)

- You can distinguish: kernel/caching/batching optimizations speed things up but do not make the model "guess more."

PHASE 13: Interpretability — systematically opening the black box

Purpose: prove that the model is analyzable (at least partly) with the right tools and approach.

P13.1 Superposition and feature geometry

- Elhage et al., "Toy Models of Superposition" (2022). (arXiv)

P13.2 Automated circuit discovery

- Conmy et al., "Towards Automated Circuit Discovery for Mechanistic Interpretability" (2023). (NIPS Papers)

Checkpoint (Phase 13)

- You can tell one concrete example of a circuit/feature and how it's traced.

BONUS: Meta-papers and philosophical debate

B1 Surveys

- Zhao et al., “A Survey of Large Language Models” (2023). (arXiv)
- Kaddour et al., “Challenges and Applications of Large Language Models” (2023). (arXiv)

B2 Philosophy & limitations

- Bender et al., “On the Dangers of Stochastic Parrots” (FAccT 2021). (ACM Digital Library)
- Kadavath et al., “Language Models (Mostly) Know What They Know” (2022). (arXiv)

Suggested learning path

Fast roadmap (1–2 months, “core papers”)

- Vaswani 2017 (Transformer) (arXiv)
- Brown 2020 (GPT-3, few-shot) (arXiv)
- Holtzman 2020 (decoding/nucleus) (arXiv)
- Ouyang 2022 (InstructGPT/RLHF) (arXiv)
- Xiao 2023 / ICLR 2024 (attention sinks) (arXiv)
- Wiher et al. 2022 (decoding taxonomy) (ACL Anthology)
- Malach 2023 (universal learners) (arXiv)

Full roadmap (3–6 months)

- Go Phase 0 → 13, prioritize: Phase 0, 3, 4, 6, 11 as the “spine.”

Hands-on roadmap (learn alongside code)

- N-gram + smoothing following Chen & Goodman (arXiv)
- Implement attention from Vaswani (minimum viable) (arXiv)
- Implement decoding (greedy/top-k/top-p/temperature) and compare against Holtzman + Wiher (arXiv)
- Do a “same prompt, temp=0” demo to see reproducibility

Evidence-based talking points (to “kill illusions”)

Argument 1: “Computing a distribution” is not “random guessing”

- Holtzman shows that changing decoding alone changes output dramatically while the model is fixed. (arXiv)
- Correct conclusion: the model computes a distribution; decoding is the knob that controls how we sample from it.

Argument 2: Training differs from inference by the feedback mechanism

- InstructGPT describes a training pipeline built on preference feedback; inference doesn’t update weights. (arXiv)

Argument 3: “Emergent” does not mean “magic”

- Emergence is observed with scale and can be analyzed mechanistically (e.g., induction heads). (Proceedings NeurIPS)

Argument 4: Hallucination is structured and can be reproducible

- Surveys and source analyses show hallucination has taxonomy/mechanisms, not meaningless randomness. (arXiv)